

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

MAY EXAMINATION

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS

MONDAY, 22 MAY 2023

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATION AND NETWORKING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING

Instruction to Candidates:

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Candidates are allowed to use non-scientific calculator.

Answer questions only in the answer booklet provided.

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS**SECTION A**

- Q1. (a) Generate a smart fish farming solution (architecture diagram) with Amazon Web Services (AWS) cloud platform. Justify the selection of each cloud service. Your design should consider:
- Deploying sensors for water parameters capturing,
 - Storing data for fish growth analysis,
 - Generating serverless functions for data processing,
 - Notifying fish farmer for abnormal pond's conditions,
 - Having data pipeline for different cloud services,
 - Storing sensor log files into the cloud storage,
 - Integrating some relevant intelligent cloud services, and
 - Visualizing sensor data for trend analysis.

(12 marks)

- (b) You have an Amazon Elastic Compute Cloud (EC2) Instance in development that interacts with the AWS DynamoDB. You will deploy the EC2 Instance in production environment soon. Which AWS service(s) should be used for a secure communication between the EC2 Instance and the DynamoDB? Justify your answer.

(3 marks)

- (c) AWS offers several compute and database options to meet different needs. Which of the AWS service(s) can be used to host an off-premise MySQL database with low admin operations? Justify your answer.

(4 marks)

- (d) Answer the following questions with short justification in the context of AWS.

(i) What purpose does Amazon SageMaker serve? (2 marks)

(ii) List out **TWO (2)** use cases for Amazon SageMaker that can help an organization in accelerating machine learning initiatives. (2 marks)

(iii) Give **TWO (2)** features of Amazon QuickSight that help in deriving business value from its data. (2 marks)

[Total : 25 marks]

- Q2. (a) Which type of persistent communication can be best applied in transferring a patient's record from their admitted hospital A to their physician's hospital B? Please elaborate on how to perform the selected communication service with the scenario above. *[Hints: Both hospitals are in different cities and aim to eliminate the traditional overhead associated with operating in-house communication infrastructures.]* (4 marks)

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS**Q2. (Continued)**

- (b) Consider a cloud infrastructure that is hosted off-premise. Suggest a **cloud deployment model** that best suits the requirements of having full control of and responsibility for the company data by internal users. Justify your selection. (4 marks)
 - (c) Explain **TWO (2)** benefits of middleware to distributed system programmers, providing an example for each benefit. (4 marks)
 - (d) Explain **TWO (2)** key differences between a network operating system and a distributed operating system, and how do these differences affect their respective use cases. (4 marks)
 - (e) What are the differences between a single-threaded server and a multi-threaded server, and what are the advantages and disadvantages of each in terms of performance, resource utilization, complexity, and scalability? (9 marks)
- [Total : 25 marks]

- Q3.
- (a) Given a scenario of **4 Generals AND 1 Traitor**, find the traitor by using Byzantine Generals algorithm. Show all the steps and diagram necessary to your answer. (11 marks)
 - (b) Under which condition the Byzantine Generals Problem cannot be solved? (3 marks)
 - (c) Suppose there are four processes *A*, *B*, *C* and *D*. All clocks run at the same rate (2 units of time) but initially *A*'s clock reads 78, *B*'s clock reads 88, *C*'s clock reads 44 and *D*'s clock reads 66. At time 78 by *A*'s clock, *A* sends a message to *B*, this message takes 6 units of time to reach *B*. *B* then waits 4 units of time and then sends a message onto *C* which takes 2 units of time to reach *C*. Later, *C* also waits 6 units of time and then sends a message onto *D*. The message reaches *D* after 2 units of time. After *D* waits for 4 units of time, a message takes another 2 units of time to reach *B* again. Assuming that the system implements Lamport's timestamps, draw a picture to illustrate the timestamps for the messages and how the timestamps are obtained. (8 marks)
 - (d) Consider a distributed system consisting of four replicated servers. Each of the servers is available at any instant with a probability of 90%. What is the overall system availability, if the system is designed so that the system can be operational if any one of the four servers is working? (3 marks)
- [Total : 25 marks]

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS**SECTION B**

- Q4. (a) From your point of view, how does ChatGPT utilize cloud computing technology to power its natural language processing capabilities? Please provide **THREE (3)** points to support your justification. (6 marks)
- (b) Can you discuss **THREE (3)** challenges that organizations might encounter when implementing a chatbot like ChatGPT? (6 marks)
- (c) Multi-Agent Systems (MAS) has shown significant impact in solving large scale distributed systems for the past few decades.
- (i) How do you define non-player characters with Artificial Intelligence (AI) in the multiplayer online battle as a distributed system? (4 marks)

Field	Value
Purpose	INFORM
Sender	max@http://fanclub-beatrix.royalty-spotters.nl:7239
Receiver	elke@iiop://royalty-watcher.uk:5623
Language	Prolog
Ontology	genealogy
Content	female(beatrix),parent(beatrix,juliana,bernhard)

Figure 4.0

- (ii) Figure 4.0 shows an example of agent communication message using FIPA ACL format. Interpret this message. (3 marks)
- (iii) Beside multiplayer online game, name **TWO (2)** more MAS applications in the real world. (2 marks)
- (d) Recommend **ONE (1)** type of distributed computer system which is suitable to support 8 classes of students (40 students per class), where different classes required different types of mathematic/computer applications. Suppose that UTAR can only hire one lab administrator to take care of the machines. Provide **TWO (2)** justifications for your selection of distributed computer system. (4 marks)

[Total : 25 marks]

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS

- Q5. (a) Generate a smart car solution (architecture diagram) with AWS cloud platform. Your design should consider gathering of various data from the car, store data into cloud storage and database for further analysis and visualization. (7 marks)
- (b) Four clocks W , X , Y and Z are performing internal synchronization by using Berkeley algorithm. W is reading 18:05:00.0 (hr:min:sec) when it is chosen as the master clock. At the same time, X , Y and Z are reading 18:16:00.0, 17:58:00.0 and 18:09:00.0, respectively. Illustrate the clock synchronization process with diagrams from initialization to adjust the time and show the calculations. (10 marks)
- (c) Answer the following questions with short justification in the context of AWS.
- (i) Michael has limited budget for hosting his distributed application. This application periodically processes large volumes of data across multiple compute instances. The application is designed to recover smoothly from compute instance failures. Which of the Amazon EC2 pricing model will meet his requirements? (2 marks)
- (ii) A company wants to utilize AWS Storage. For the company:
- low storage cost is paramount,
 - the data is rarely retrieved, and
 - data retrieval times of several hours are acceptable.
- What is the best AWS storage option to use? (2 marks)
- (iii) What is the best way to back up an Amazon Elastic Block Store (EBS) volume in AWS? (2 marks)
- (iv) Which AWS security feature is associated with a Subnet in an Amazon Virtual Private Cloud to protect against incoming traffic requests? (2 marks)
- [Total : 25 marks]
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